

Quadratic functions - Set 2 - Solutions

Specification MM1.3, M4.5, M4.11-M4.14.

Answer key

Question	Answer	Topic
1	A	Quadratic functions
2	D	Quadratic functions
3	A	Quadratic functions
4	B	Quadratic functions
5	C	Quadratic functions
6	E	Quadratic functions
7	E	Quadratic functions
8	H	Quadratic functions
9	E	Quadratic functions
10	B	Quadratic functions
11	F	Quadratic functions
12	F	Quadratic functions

Worked solutions

Question 1 QUA-18 · Quadratic functions · spec MM1.3

Find the range of values of x for which $x^2 - 7x + 10 < 0$ and $x^2 - 4 > 0$.

Answer: A

$(x - 2)(x - 5) < 0$ gives $2 < x < 5$; $x^2 > 4$ gives $x < -2$ or $x > 2$. The intersection is $2 < x < 5$.

Full method - how and why

Two simultaneous quadratic inequalities. Solve each separately, then intersect on a number line.

- 1 First: $x^2 - 7x + 10 < 0$ factorises as $(x - 2)(x - 5) < 0$, giving $2 < x < 5$.

why: An upward parabola is **below** the axis between its roots.

- 2 Second: $x^2 - 4 > 0$ gives $x < -2$ or $x > 2$.

why: Above the axis means outside the roots - two intervals.

- 3 Intersect: $(2, 5)$ overlaps $(2, \infty)$ entirely and misses $(-\infty, -2)$ completely, so the answer is $2 < x < 5$.

why: 'And' means intersection: a value must satisfy both, so draw both solution sets on one line and take the common part.

Where it goes wrong

- Taking the union and including $x < -2$ - the word is 'and', not 'or'.
- Getting the first inequality backwards ($x < 2$ or $x > 5$) by forgetting that < 0 means between the roots.

Takeaway < 0 between the roots, > 0 outside them (for an upward parabola); then combine solution sets on a number line.

Question 2 QUA-19 · Quadratic functions · spec MM1.3

The expression $x^2 + 6x + c$ can be written as $(x + p)^2 + q$. Given that $q = -c$, find the value of c .

Answer: D

$(x + 3)^2 + (c - 9)$, so $q = c - 9 = -c$, giving $2c = 9$ and $c = \frac{9}{2}$.

Full method - how and why

Completing the square symbolically, then imposing a relationship between the two constants. It is an equation in c , not in x .

- 1 Complete the square: $x^2 + 6x + c = (x + 3)^2 - 9 + c$.

why: Half of 6 is 3; subtracting 9 compensates for the +9 the square introduces.

- 2 Identify $p = 3$ and $q = c - 9$.

why: Compare with the required form $(x + p)^2 + q$ term by term.

- 3 Impose $q = -c$: $c - 9 = -c$.

why: The condition given in the question is a link between q and c , so substitute your expression for q .

- 4 Solve: $2c = 9$, so $c = \frac{9}{2}$.

why: A simple linear equation - just collect the c terms.

Where it goes wrong

- Writing $q = c + 9$ (sign error) and getting $c = -\frac{9}{2}$.
- Solving $c - 9 = c$ and concluding there is no solution.

Takeaway Complete the square with the constant left as a letter; the extra condition then becomes an ordinary equation in that letter.

Question 3 QUA-21 · Quadratic functions · spec MM1.3

The parabola $y = x^2 - 4x + 3$ is reflected in the x -axis and then translated 2 units in the positive y -direction. What is the maximum value of the resulting function?

Answer: A

$x^2 - 4x + 3 = (x - 2)^2 - 1$, minimum -1 . Reflecting gives maximum 1 ; translating up by 2 gives maximum 3 .

Full method - how and why

Transformations applied to a quadratic. Track what happens to the **vertex** rather than to the whole equation.

- 1 Complete the square: $x^2 - 4x + 3 = (x - 2)^2 - 1$, so the minimum is -1 at $x = 2$.

why: The turning point is the only feature the transformations act on in a way that matters.

- 2 Reflect in the x -axis: $y \rightarrow -y$, so the minimum -1 becomes a **maximum** of 1 .

why: Reflection in the x -axis flips the sign of every y -value, turning a minimum into a maximum.

- 3 Translate 2 up: the maximum becomes $1 + 2 = 3$.

why: A vertical translation adds to every y -value, including the extreme one.

Where it goes wrong

- Forgetting that the reflection converts the minimum into a maximum and reporting $-1 + 2 = 1$.
- Reflecting the x -coordinate as well - the turning point stays at $x = 2$ throughout.

Takeaway For transformation questions, follow one key point (usually the vertex) through each step in order.

Question 4 QUA-23 · Quadratic functions · spec MM1.3

The line $y = mx$ meets the curve $y = x^2 + 9$ at two distinct points. Find the complete set of values of m .

Answer: B

$x^2 - mx + 9 = 0$ needs $m^2 - 36 > 0$, so $m < -6$ or $m > 6$.

Full method - how and why

A line meeting a curve twice: substitute and demand a positive discriminant.

- 1 Substitute $y = mx$ into $y = x^2 + 9$: $mx = x^2 + 9$.

why: Intersections are the common solutions, found by equating the two expressions for y .

- 2 Rearrange: $x^2 - mx + 9 = 0$.

why: Standard form is needed before the discriminant can be read off.

- 3 Two distinct intersections $\Rightarrow$ discriminant > 0 : $m^2 - 36 > 0$.

why: Here $b = -m$, and $b^2 = m^2$ regardless of sign.

- 4 Solve $m^2 > 36$: $m < -6$ or $m > 6$.

why: Two intervals, since $m^2 > 36$ means $|m| > 6$; geometrically, steep lines in either direction cut the parabola twice.

Where it goes wrong

- Answering $-6 < m < 6$ (the region where the line misses the parabola entirely).
- Writing the discriminant as $-m^2 - 36$ by mishandling the sign of b .

Takeaway b^2 kills the sign of b : conditions like this are almost always symmetric in $\pm m$.

Question 5 QUA-24 · Quadratic functions · spec MM1.3

Let $f(x) = x^2 - 2x - 8$. For how many integers n is $f(n) < 0$?

Answer: C

$(x - 4)(x + 2) < 0$ gives $-2 < x < 4$; the integers are $-1, 0, 1, 2, 3$: five of them.

Full method - how and why

Counting **integers** in the region where a quadratic is negative. Solve the inequality, then count carefully at the ends.

- 1 Factorise: $n^2 - 2n - 8 = (n - 4)(n + 2)$.

why: Two numbers multiplying to -8 and adding to -2 : -4 and $+2$.

- 2 Negative between the roots: $-2 < n < 4$.

why: Upward parabola, so $f < 0$ strictly between the roots.

- 3 List the integers strictly inside: $-1, 0, 1, 2, 3$ - five of them.

why: The endpoints -2 and 4 give $f = 0$, which is **not** less than zero, so they are excluded; and 0 is easy to forget.

Where it goes wrong

- Including -2 and 4 and answering 7.
- Forgetting 0 is an integer and answering 4.

Takeaway After solving an inequality, list the integers explicitly - endpoints and zero are the two things people drop.

Question 6 X2-30 · Quadratic functions · spec MM1.3

The equation $x^2 + bx + c = 0$ has real coefficients. Which of the following is a **sufficient** condition for it to have two distinct real roots?

Answer: E

$c < 0$ makes the discriminant $b^2 - 4c$ positive. $b^2 \geq 4c$ allows equality (a repeated root); the others do not control the discriminant.

Full method - how and why

'Sufficient' means the condition alone guarantees the conclusion. Translate into the discriminant and check each option **guarantees** positivity.

- 1 Two distinct real roots requires discriminant $b^2 - 4c > 0$ (strictly).

why: Zero discriminant gives a repeated root, not two distinct ones.

- 2 If $c < 0$ then $-4c > 0$, so $b^2 - 4c \geq -4c > 0$. **Guaranteed** - sufficient.

why: $b^2 \geq 0$ whatever b is, so the whole expression is positive.

- 3 $b^2 \geq 4c$ allows equality, which gives a repeated root - not sufficient for **distinct** roots.

why: Read the word 'distinct'; the boundary case ruins it.

- 4 $b \neq 0, c > 0, b > 0$ give no control: e.g. $b = 1, c = 1$ has discriminant $-3 < 0$, no real roots.

why: One failing example shows a condition is not sufficient.

Where it goes wrong

- Choosing $b^2 \geq 4c$ because it looks like the discriminant condition - it is the condition for **at least one** real root.
- Confusing 'sufficient' with 'necessary': $c < 0$ is not necessary ($x^2 - 5x + 6$ has two distinct roots with $c = 6$).

Takeaway Distinct real roots $\Leftrightarrow b^2 - 4c > 0$. A sufficient condition must force that strictly, with no exceptions.

Question 7 QUA-04 · Quadratic functions · spec MM1.3

The roots of $x^2 - 5x + 2 = 0$ are α and β . Find the value of $\alpha^2 + \beta^2$.

Answer: E

$$\alpha + \beta = 5 \text{ and } \alpha\beta = 2. \text{ So } \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 25 - 4 = 21.$$

Full method - how and why

A symmetric function of the roots. The roots themselves are irrational, so use the sum and product rather than the quadratic formula.

- 1 Read off $\alpha + \beta = -\frac{b}{a} = 5$ and $\alpha\beta = \frac{c}{a} = 2$.

why: These come straight from the coefficients; you never need the actual roots.

- 2 Use the identity $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$.

why: Expanding $(\alpha + \beta)^2$ gives $\alpha^2 + 2\alpha\beta + \beta^2$, so subtracting $2\alpha\beta$ leaves the sum of squares.

- 3 Substitute: $25 - 2(2) = 21$.

why: Arithmetic only.

Where it goes wrong

- Using $(\alpha + \beta)^2 = \alpha^2 + \beta^2$ and answering 25.
- Subtracting $\alpha\beta$ once instead of twice, giving 23 - both wrong values are offered.

Takeaway Any symmetric expression in the roots can be built from $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$; the quadratic formula is rarely needed.

Question 8 QUA-06 · Quadratic functions · spec MM1.3

The function $f(x) = x^2 - 6x + 13$ is defined for all real x . Which of the following statements is/are true?

I $f(x) > 0$ for all real x

II The graph of $y = f(x)$ is symmetric about the line $x = 3$

III The equation $f(x) = 4$ has exactly one real solution

Answer: H

$f(x) = (x - 3)^2 + 4 \geq 4 > 0$, so I is true. The axis of symmetry is $x = 3$, so II is true. $f(x) = 4$ only when $(x - 3)^2 = 0$, i.e. $x = 3$: exactly one solution, so III is true.

Full method - how and why

Three statements about one quadratic. Completing the square once answers all three at a stroke.

- 1 Complete the square: $x^2 - 6x + 13 = (x - 3)^2 + 4$.

why: This single form contains the range, the line of symmetry and the vertex - so do it first, before looking at the statements.

- 2 I: $(x - 3)^2 \geq 0$ so $f(x) \geq 4 > 0$ for every x . **True.**

why: The minimum value is 4, comfortably positive; equivalently the discriminant $36 - 52 < 0$, so the curve never reaches the axis.

- 3 II: the vertex is at $x = 3$, and a parabola is symmetric about the vertical line through its vertex. **True.**

why: Symmetry about $x = h$ is immediate from the $(x - h)^2$ form.

- 4 III: $f(x) = 4$ means $(x - 3)^2 = 0$, so $x = 3$ only - exactly one solution. **True.**

why: A horizontal line through the vertex meets the parabola once; any higher line meets it twice.

- 5 All three hold, so the answer is 'I, II and III'.

why: Decide each statement independently, then match to the option list.

Where it goes wrong

- Assuming $f(x) = 4$ has two solutions because quadratics 'usually' do - $y = 4$ is the tangent line at the vertex.
- Reading the symmetry line off as $x = -3$.

Takeaway Completing the square answers questions about sign, range, symmetry and repeated roots all at once.

Question 9 QUA-09 · Quadratic functions · spec MM1.3

Find the sum of all real values of x satisfying $(x^2 - 3x)^2 - 2(x^2 - 3x) - 8 = 0$.

Answer: E

Let $u = x^2 - 3x$: $(u - 4)(u + 2) = 0$. $x^2 - 3x - 4 = 0$ gives $x = 4, -1$ (sum 3). $x^2 - 3x + 2 = 0$ gives $x = 1, 2$ (sum 3). Total 6.

Full method - how and why

A quadratic in the expression $x^2 - 3x$. After substituting back you get two quadratics, and the question only wants the total of all four roots.

1 Let $u = x^2 - 3x$; the equation becomes $u^2 - 2u - 8 = 0$.

why: The same expression appears squared and linearly, which is the signal for a substitution.

2 Factorise: $(u - 4)(u + 2) = 0$, so $u = 4$ or $u = -2$.

why: Both values are usable - neither is ruled out, because $x^2 - 3x$ can take any value $\geq -\frac{9}{4}$.

3 Case $u = 4$: $x^2 - 3x - 4 = 0$. Its roots sum to 3 (sum $= -b/a$). Discriminant $9 + 16 > 0$, so both are real.

why: You never need the roots themselves - the sum of roots comes straight from the coefficients.

4 Case $u = -2$: $x^2 - 3x + 2 = 0$. Roots sum to 3; discriminant $9 - 8 > 0$, so both are real.

why: Check the discriminant in each case: a case with no real roots would contribute nothing to the total.

5 Total: $3 + 3 = 6$.

why: Four real values of x , summing to 6.

Where it goes wrong

- Answering 3 by solving only one of the two cases.
- Adding the u values ($4 + (-2) = 2$) instead of the x values.

Takeaway With a substitution, each value of u gives its own equation in x ; check each for real roots and use 'sum of roots $= -b/a$ ' to avoid solving.

Question 10 QUA-14 · Quadratic functions · spec MM1.3

The equation $x^2 - 2mx + m + 6 = 0$ has two distinct positive roots. Find the complete set of values of m .

Answer: B

Distinct real roots: $4m^2 - 4(m + 6) > 0 \Rightarrow m^2 - m - 6 > 0 \Rightarrow (m - 3)(m + 2) > 0$, so $m < -2$ or $m > 3$. Both roots positive needs sum $2m > 0$ and product $m + 6 > 0$, so $m > 0$. Combined: $m > 3$.

Full method - how and why

'Two distinct **positive** roots' is three conditions at once: discriminant positive, sum of roots positive, product of roots positive. All three must hold.

1 Discriminant: $4m^2 - 4(m + 6) > 0 \Rightarrow m^2 - m - 6 > 0 \Rightarrow (m - 3)(m + 2) > 0$, so $m < -2$ or $m > 3$.

why: Distinctness and reality come from $b^2 - 4ac > 0$; this alone allows negative m .

2 Sum of roots $= 2m > 0$, so $m > 0$.

why: If both roots are positive their sum is positive. This is the condition that kills the $m < -2$ branch.

3 Product of roots $= m + 6 > 0$, so $m > -6$.

why: If both roots are positive their product is positive; this also rules out one positive and one negative root.

4 Intersect all three: ($m < -2$ or $m > 3$) and $m > 0$ and $m > -6$ gives $m > 3$.

why: Take the overlap on a number line - the answer must satisfy every condition simultaneously.

Where it goes wrong

- Stopping after the discriminant and choosing $m < -2$ or $m > 3$ - which is offered.
- Forgetting that sum **and** product must both be positive: either one alone is not enough.

Takeaway Both roots positive $\Leftrightarrow$ discriminant ≥ 0 , sum > 0 and product > 0 ; sign conditions on roots always come in this package.

Question 11 QUA-17 · Quadratic functions · spec MM1.3

The quadratic equation $ax^2 + bx + c = 0$ (with $a \neq 0$) has real coefficients. Which of the following statements is/are always true?

- I If $ac < 0$, the equation has two distinct real roots.
- II If $b = 0$, the equation has no real roots.
- III If the equation has a repeated root, then $b^2 = 4ac$.

Answer: F

I: $b^2 - 4ac > 0$ whenever $ac < 0$, so true. II: $x^2 - 4 = 0$ has $b = 0$ and real roots, so false. III: a repeated root means the discriminant is zero, so true.

Full method - how and why

Three general claims about $ax^2 + bx + c = 0$. Each is a discriminant statement in disguise; the false one needs a counterexample.

- 1 I: if $ac < 0$ then $-4ac > 0$, so $b^2 - 4ac \geq -4ac > 0$: two distinct real roots. **True.**
why: $b^2 \geq 0$ always, so a positive $-4ac$ forces the whole discriminant positive. Geometrically, a and c of opposite signs means the curve crosses the axis.
- 2 II: test $b = 0$ with $x^2 - 4 = 0$, which has roots ± 2 . **False.**
why: One counterexample settles a 'for all' claim. With $b = 0$ the discriminant is $-4ac$, whose sign depends on a and c .
- 3 III: a repeated root means the discriminant is zero, i.e. $b^2 - 4ac = 0$, i.e. $b^2 = 4ac$. **True.**
why: This is the definition of the repeated-root case, just rearranged.
- 4 I and III only.
why: Match the verdicts to the options.

Where it goes wrong

- Believing II because $b = 0$ 'removes' the linear term and the graph 'sits on the axis' - it does not; $x^2 - 4$ is a counterexample.
- Mis-stating III as $b^2 = -4ac$.

Takeaway Statements about roots translate into statements about $b^2 - 4ac$; to refute one, look for the simplest numbers that fit the hypothesis.

Question 12 QUA-20 · Quadratic functions · spec MM1.3

For how many integer values of k does $x^2 + kx + 12 = 0$ have integer roots?

Answer: F

Integer roots r, s with $rs = 12$: pairs $(1, 12), (2, 6), (3, 4)$ and their negatives, giving $k = -(r + s) = -13, -8, -7, 13, 8, 7$: six values.

Full method - how and why

Integer roots means an integer factorisation of the constant term. Count systematically and do not forget negative pairs.

- 1 If the roots are integers r and s then $rs = 12$ and $r + s = -k$.
why: For a monic quadratic, product of roots $= c$ and sum $= -b$.
- 2 List the positive factor pairs of 12: $(1, 12), (2, 6), (3, 4)$.
why: Work through divisors in order so none is missed.
- 3 Each pair also has a negative version $(-1, -12), (-2, -6), (-3, -4)$, since two negatives also multiply to $+12$.
why: The product is positive, so the roots share a sign - both positive or both negative.
- 4 Each of the six pairs gives a different $k = -(r + s)$: $-13, -8, -7, 13, 8, 7$. So six values.
why: Check they are distinct before counting.

Where it goes wrong

- Listing only the positive pairs and answering 3.
- Counting $(1, 12)$ and $(12, 1)$ as different, which double-counts.

Takeaway Integer-root problems become divisor-counting problems; the sign of the product tells you whether the roots share a sign.